

The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE
A LEVEL 01 MODULE, AUTUMN SEMESTER 2020-2021
DATABASES AND INTERFACES
COMP1044

Time allowed: **TWO Hours**

Answer ALL questions

ADDITIONAL MATERIAL: Nil.

INSTRUCTION FOR STUDENTS:

- The university's honour code applies to this individual short time assessment. The submitted answers **must be your own work** where **answers are attempted without any help from others**. University academic misconduct rules applies here in the case of plagiarism, collusion or false authorship.
- Write the name and student ID at the top of your answer sheets.
- Students are not allowed to ask questions during the short time assessment in accordance with the no shout-out policy.
- You may answer the questions in any order but write the question numbers correctly and clearly.
- At the end of the exam, **upload your** solutions and name the file as **StudentID**. Upload your solution file onto **Moodle**.
- Please ensure that you upload the correct files to the folder labelled as "Final examination submission folder" in Moodle. Request for resubmission will not be accepted after the submission.
- Students should stop writing the answers at the end of exam time. Answer scripts must be submitted to Moodle within **thirty minutes** after the end of exam. In case of any technical issues, the students need to contact the module convener in this period.

IMPORTANT: Late submission will not be accepted.

Question 1**[20 marks]**

- (a) The tables below are used by a company to record data about its employees and departments.

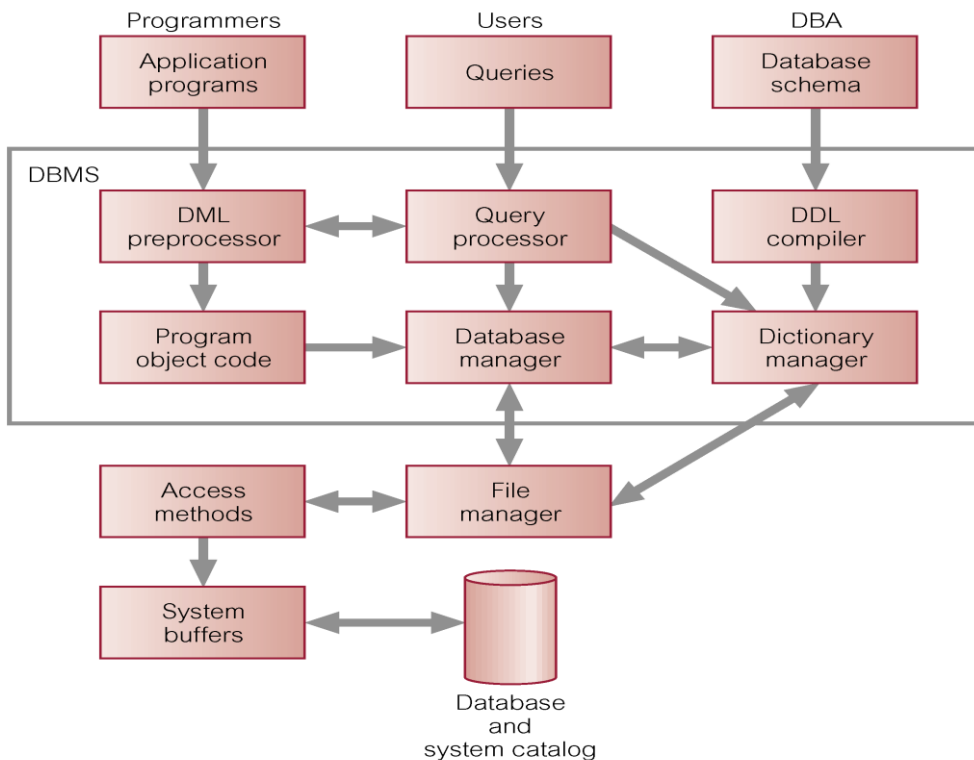
Table name: EMPLOYEE							
EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	17-DEC-90	800.00		20
7499	ALLEN	SALESMAN	7698	20-FEB-91	1600.00	300.00	30
7844	TURNER	SALESMAN	7698	08-SEP-91	1500.00	0.00	30
7876	ADAMS	CLERK	7788	23-SEP-91	1100.00		20
7900	JAMES	CLERK	7698	03-DEC-91	950.00		30

Table name: DEPARTMENT		
DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

Formulate SQL statements for the following queries:

- (i) Display the highest salary earned by an analyst [3 marks]
- (ii) Increase the salary of all clerks by 10%. [2 marks]
- (iii) List the names of all managers and the departments (**ENAME** and **DNAME**) they manage. [2 marks]
- (iv) List the department name and the total number of employees in each department. [4 marks]
- (v) List the **ENAME** of clerks who are paid a higher salary than the clerk named 'James'. Use the 'sub query' concept. [4 marks]
- (b) Critically analyse how the FIVE (5) functional components contribute to a Database Management System as shown in the diagram below: [5 marks]

Turn Over



Question 2

[20 marks]

- (a) The IT Training Group (Kuala Lumpur) has contacted you to create a conceptual model by using the Entity–Relationship data model for a database that will meet the information needs for its training programme. The Company Director gives you the following description of the training group’s operating environment.

The Company has twelve instructors and can handle up to one hundred trainees per training session. The Company offers five advanced technology courses, each of which is taught by a teaching team of two or more instructors. Each instructor is assigned to a maximum of two teaching teams or may be assigned to do research in a particular field. Each trainee undertakes one advanced technology course per training session.

From the enterprise rules above, draw an Entity Relationship (ER) diagram, with appropriate attributes, keys, cardinalities and participation constraints.

[10 marks]

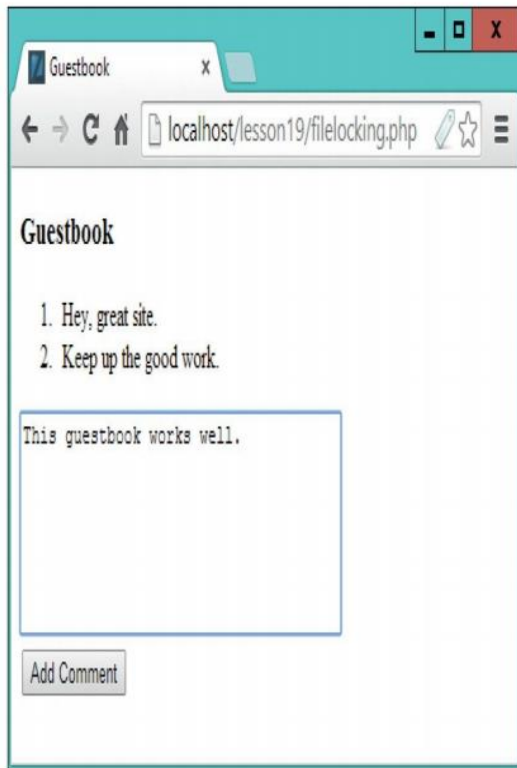
- (b) Describe **FIVE (5)** main characteristics of the database approach and contrast it with the file-based approach.

[10 marks]

Turn Over

Question 3**[20 marks]**

- (a) Identify with example THREE (3) different types of errors in PHP. [6 marks]
- (b) Develop the following output using php code as provided in figure below:



[6 marks]

- (c) List out the different ways an HTML elements can be accessed in a JScript code.
- Gets an element by its ID name
 - Gets all the elements that have the given classname.
 - Gets all the elements that have the given tag name.
 - This function takes css style selector and returns the first selected element.
- [8 marks]

Question 4**[20 marks]**

- (a) Normalize the below Employee_Project table into Third Normal Form (3NF).

Project_No	Project_Name	Emp_No	Emp_Name	Hour	Chg_Hour	Job_Class
CR76	Payroll	PG4	Willy	35	85.00	Programmer
		PG16	Smith	45	78.00	DBA
CR56	Barcode	PG4	Willy	50	85.00	Programmer
		PG36	Kamal	37	65.00	Analyst
		PG16	Smith	45	78.00	DBA

[10 marks]

Turn Over

- (b) Establishing referential integrity using SQL: 2012 compliant database, explain the differences among the ON UPDATE RESTRICT, ON UPDATE CASCADE, and ON UPDATE SET NULL clauses, and results of declaring ON DELETE CASCADE. [10 marks]

Question 5**[20 marks]**

- (a) Employees and other insiders present the greatest database security problems to companies. Explain with suitable examples, **FIVE (5)** potential threats posed by these personnel. [10 marks]
- (b) An online bookstore will require additional security measures to protect its database transactions as opposed to one which does not have an online presence. Describe any **TWO (2)** of such measures and explain how those measures can safeguard the bookstore's database. [6 marks]
- (c) Briefly explain with example the following keys: [4 marks]
- i. *candidate keys*
 - ii. *primary key*

End of Exam